Learning Transferable Discriminative Knowledge From Attribute-Aligned Hyperspectral Images

Qichao Liu, Liang Xiao, *Member, IEEE*, Nan Huang, and Jinhui Tang, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) classification faces the inherent challenge of small sample learning, primarily due to the difficulty in labeling vast land covers. Meta-learning, with its ability to learn transferable meta-knowledge from existing HSIs, is seen as a promising solution. However, different HSIs usually have varying distributions manifested as differing spectral wavelengths and reflectance shifts, which is often neglected in existing methods, stalling the acquisition of transferable features. To address this issue, we introduce an attribute-driven spectral alignment (ADSA) method, which parameterizes and embeds spectral attributes (i.e., spectral wavelengths and reflectance shifts) into a domain-adaptation model, aiming to decouple the domain-specific attributes of different HSIs in an unsupervised manner. After training, the parameterized attributes of source-domain (SD) HSIs can be substituted with those of the target domain (TD), allowing the decoder of ADSA to rebuild new HSIs sharing identical spectral attributes. By this means, numerous distribution-consistent labeled samples preserving inherent spectral–spatial structures can be obtained. Then, a 3-D residual prototypical network (RPN) based on 3-D convolutions and metric learning is designed to model complex structures of HSIs, which in combination with the few-shot learning (FSL) framework can extract valuable discriminative knowledge from these auxiliary samples. Finally, by applying this learned knowledge to the TD HSI, only a small number of labeled samples are required to obtain satisfactory performance. Extensive experiments on four real-world HSIs demonstrate the effectiveness of our method, and the performance outperforms several state-of-the-art methods.**

***Index Terms*— Attribute-driven spectral alignment (ADSA), cross domain, few-shot learning (FSL), hyperspectral image (HSI) classification, meta-learning.**

I. Introduction

REMOTE sensing hyperspectral images (HSIs) are 3-D cubic data containing rich radiometric, spatial, and spectral information, where each pixel is a spectrum recording the absorption characteristics of land cover at different wavelengths of electromagnetic waves in a range of 300–2500 nm. Benefiting from this property, HSIs are capable of distinguishing substances at pixel level, having been widely applied to many industrial fields, such as environmental monitoring [1], resource exploration [2], and agricultural assessment [3]. As a fundamental task in these applications, accurately classifying each spectrum determines its reliability and availability, which is often called HSI classification.

In early studies, statistics-based approaches, such as support vector machine (SVM) [4], random forest [5], and nonlinear regression [6], were commonly used for HSI classification due to their simplicity and practicality. Then, representation-based learning methods, such as sparse representation [7], [8], low-rank representation [9], and collaborative representation [10], were later introduced to reduce the spectral complexity and improve feature learning robustness. Owing to the influence of unregulated dark currents in hyperspectral sensors, HSIs are often impacted by intricate noise, leading to a substantial prevalence of outliers within the resulting classification maps. Consequently, researchers have integrated spatial information from HSIs into the aforementioned methods, resulting in spectral–spatial classification frameworks [11], [12], [13], [14] that significantly enhanced classification accuracies.

Conventional approaches predominantly depend on manual priors and often necessitate distinct empirical parameters for different HSIs, leading to suboptimal robustness and classification performance. In contrast, deep learning techniques have garnered widespread adoption in HSI classification tasks in recent years, owing to their capacity to automatically learn adaptive features from samples. Spanning 1-D [15], 2-D [16], and 3-D convolutional neural networks (CNNs) [17], [18] as well as residual [19], dense [20], attentive [21], [22], [23], and graph [24] neural networks, deep learning has demonstrated substantial advancements within the field of hyperspectral remote sensing. However, most of these methods require massive labeled samples to achieve expected performance, which is generally difficult to achieve in practice due to limitations in human and material resources.

To mitigate the challenge posed by limited labeled samples, metric learning methods [25], [26], [27], [28], [29], [30], [31] have been employed to enhance small-sample learning ability by explicitly optimizing the decision boundaries of models. But, they can only learn knowledge from a few given supervised samples in a single data domain, leading to limited improvement. In contrast, transfer learning [32], [33] aims to learn prior knowledge from source-domain (SD) HSIs and

Manuscript received 15 August 2023; revised 20 October 2023; accepted 6 November 2023. Date of publication 8 November 2023; date of current version 22 November 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61871226 and Grant 61925204, in part by the Jiangsu Provincial Nature Science Foundations of China under Grant BK20230914, in part by the Jiangsu Funding Program for Excellent Postdoctoral Talent under Grant 2022ZB280, in part by the China Postdoctoral Science Foundation Funded Project under Grant 2022M721627, in part by the Fundamental Research Funds for the Central Universities under Grant JSGP202203 and Grant JSGP202204, and in part by the Jiangsu Geological Bureau Research Project under Grant 2023KY11. *(Corresponding author: Liang Xiao.)*

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Digital Object Identifier 10.1109/TGRS.2023.3331244

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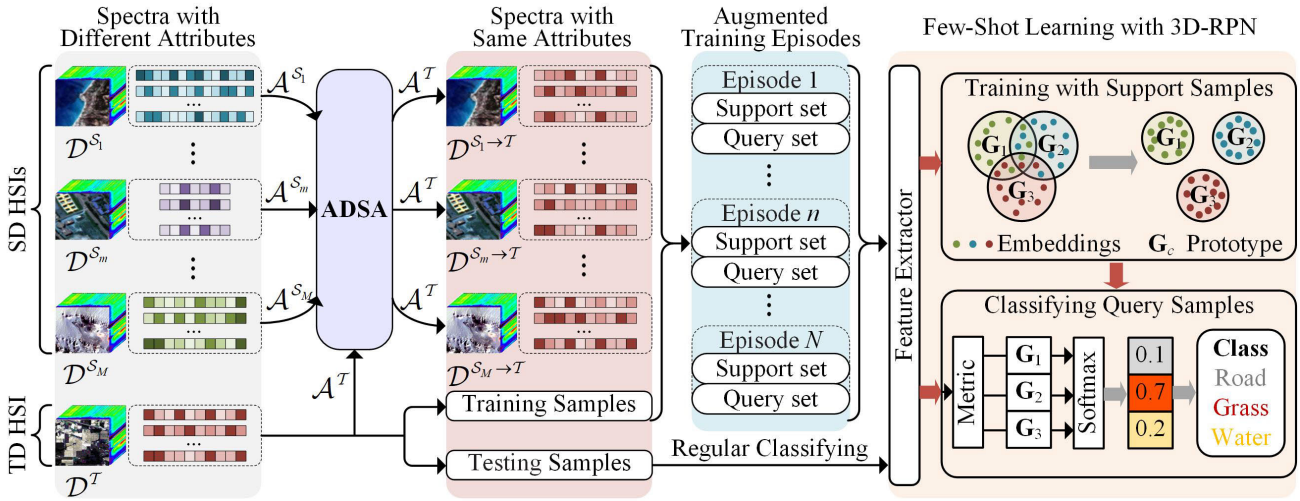


Fig. 1. Flowchart of ADSA-FSL, where M SD HSIs $\mathcal{D}^{\mathcal{S}_1}, \ldots, \mathcal{D}^{\mathcal{S}_M}$ and one TD HSI $\mathcal{D}^{\mathcal{T}}$ are fed into ADSA to learn their respective attributes $\mathcal{A}^{\mathcal{S}_1}, \ldots, \mathcal{A}^{\mathcal{S}_M}$ and $\mathcal{A}^{\mathcal{T}}$, which are then used to rebuild the new SD HSIs $\mathcal{D}^{\mathcal{S}_1 \to \mathcal{T}}, \ldots, \mathcal{D}^{\mathcal{S}_M \to \mathcal{T}}$ with the same distribution as TD HSI. Subsequently, a total of N episodes are sampled from the aligned SD and TD HSIs for training. Finally, a 3-D RPN is combined with FSL framework to learn discriminative prior knowledge from those aligned SD HSIs and leverage it to help with the small-sample classification of TD HSI.

use it to boost the classification performance of target-domain (TD) HSIs. However, the spectral spaces of different HSIs are usually not consistent, such as different spectral wavelengths, dimensionalities, and spectral reflectance shifts, making it difficult for such approaches to transfer knowledge between heterogeneous HSIs directly. To tackle this issue, domain-adaptation techniques [34], [35], [36] have been introduced to reduce the distribution gap between different HSIs in a common feature space, but they require the label spaces of the SD and TD samples to be the same. Recently, meta-learning techniques [37], [38], [39] have received increasing attention from researchers due to their ability to derive transferable meta-knowledge from SD samples, even when the label spaces are different. These methods primarily focus on reducing the statistical distribution gap between different HSIs but rarely consider the inherent spectral attributes.

In practice, different HSIs usually have varying wavelength sequences, which means that the physical meanings of their imagery bands do not correspond one-to-one, and even the number of bands may differ. Thus, it is difficult to use a single feature extractor to learn discriminative knowledge directly. In addition, domain-adaptation-based methods may lose the inherent structures of spectra when projecting HSIs into a common feature subspace, leading to the inability to fully exploit spectral structure knowledge. To address these issues, inspired by Xu et al. [40], this work proposes an attribute-driven spectral alignment (ADSA) model that aims to eliminate spectral space differences between heterogeneous HSIs while preserving their inherent structures. The ADSA model leverages adversarial learning to reduce the distribution gap between HSIs and embeds the spectral wavelengths to learn the relationship between each spectral value and its corresponding wavelength. Learnable parameters are assigned to each HSI to model their spectral reflectance shifts, which are also embedded into the domain-adaptation model for automatic learning. As a result, the spectral spaces of different HSIs can be aligned without destroying their inherent spectral and spatial structures by uniformizing their parameterized attributes in ADSA. Subsequently, a large number of labeled samples from different SD HSIs with the same distribution as the TD HSI can be acquired. Furthermore, we design a 3-D residual prototypical network (RPN) by leveraging the advantages of 3-D convolutions and metric learning to better model complex structures of HSIs, which in combination with the meta-learning forms a small-sample learning framework for HSI classification, named ADSA-FSL, as shown in Fig. 1. Extensive experiments demonstrate that our method can effectively exploit transferable discriminative knowledge from SD HSIs, with which TD HSI can achieve satisfactory performance with a small number of labeled samples.

The main contributions of this work are as follows.

1) We propose the ADSA to decouple the domain-specific attributes of different HSIs in an unsupervised manner. It can be used to mitigate the distribution gap between heterogeneous HSIs without destroying their inherent spectral and spatial structures, thereby preserving the discriminative information of the original HSIs.
2) A 3-D RPN based on 3-D convolutions and metric learning is designed to model complex structures of HSIs and discover transferable knowledge. Since the local geometric structures of spectra have interpretable physical meanings, the learned structure features using 3-D RPN possess better transferability and generalizability.
3) With ADSA and 3-D RPN, we build the ADSA-FLS framework for small-sample classification of HSIs. The ADSA can generate numerous attribute-aligned labeled HSI samples. By further combining 3-D RPN with meta-learning, ADSA-FLS can leverage prior knowledge from the SD HSIs to further boost classification performance of the TD HSI.

The rest of this article is organized as follows. Section II introduces related works. Section III describes the methodology. Section IV exhibits the comparison classification results.

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Section V further discusses and analyzes the mechanism of the proposed method. Section VI concludes this work.

II. Related Work

A. Metric Learning-Based Methods

Owing to the ability to explicitly optimize the inter- and intraclass distances of samples in the feature space, metric learning has been proven effective in improving the decision boundaries of models and small-sample learning performance. Typically, metric-learning-based few-shot learning (FSL) methods, such as prototypical networks [25], [26], [27], [28], [29] and relation networks [30], [31], have been applied to HSI classification under limited samples. However, these methods can only learn knowledge from few given supervised samples in a single data domain, leading to inadequate training of the deep models and limited performance improvement.

B. Transfer Learning-Based Methods

To increase the sources of supervised information, researchers have tried to learn prior knowledge from other SD HSIs and then use it to boost the classification performance of the TD HSI. Of these, pretraining-fine-tuning-based approaches are the most common. Yang et al. [32] proposed a transfer learning-based deep network for HSI classification, in which the prior knowledge in the SD data can be retained in the pretrained network parameters and then applied to the TD HSI by fine-tuning. He and Chen [33] proposed a transfer learning-based ensemble CNN, in which the network was first pretrained on large-scale datasets and then integrated into an ensemble system with an improved label smoothing technique to achieve enhanced classification performance. However, the spectral spaces of different HSIs are usually not consistent, such as different spectral wavelengths, dimensionalities, and spectral reflectance shifts, making it difficult for such approaches to transfer knowledge between heterogeneous HSIs directly.

C. Domain-Adaptation-Based Methods

To address the inconsistency in the distribution of SD and TD samples, researchers have introduced domain-adaptation technique to eliminate inconsistencies between heterogeneous HSIs by reducing their distribution gap in a learned common feature space. For instance, Zhang et al. [34] proposed a discriminative cooperative alignment method for cross-scene HSI classification, which addresses the challenges of domain gaps by simultaneously reducing geometric and statistical shifts in a latent subspace. Based on this, they further proposed a topological structure and semantic information transfer network [35] by combining a CNN and a graph convolutional network (GCN) to simultaneously align the semantic information and topological relationship between the SD and TD HSIs in a common subspace. Yu et al. [36] proposed an unsupervised domain-adaptation approach with contentwise alignment for HSI classification, which reduces domain gaps by incorporating class-level and style-perceive-level regularized alignment within an adversarial framework, resulting in improved performance. However, such methods require that the SD samples and the TD samples be in the same label space, i.e., the class information of the TD samples must be consistent with that of the SD samples, which means they are incapable of addressing new unseen classes, limiting their application in real scenarios.

D. Meta-Learning-Based Methods

To enable the models to handle the unseen classes, researchers introduce meta-learning techniques to learn transferable meta-knowledge from SD HSIs and then use it to improve the classification performance of TD HSI, allowing the label spaces of the source and target HSIs to be different. For instance, Li et al. [37] proposed a deep cross-domain FSL (DCFSL) method by unifying FSL and domain adaptation, which used a conditional adversarial domain-adaptation strategy to align distribution and executed FSL in both SD and TD samples simultaneously. Similarly, Zhang et al. [38] proposed a graph information aggregation cross-domain FSL (GiaCFSL) framework, in which both local and nonlocal spatial information was utilized to mitigate the impact of domain shift. Moreover, Wang et al. [39] proposed a classwise attentive cross-domain FSL (CACFSL) framework, which further improved interclass decision boundaries and enhanced domain-invariant representation extraction by employing a classwise attention metric mechanism and an asymmetric-domain adversarial learning strategy. By combining meta-learning with domain adaptation, those class-agnostic methods can discover transferable meta-knowledge from SD data and learn discriminative embedding models for the TD HSI, without being affected by the inconsistency of class information between SD and TD samples.

III. Proposed Method

To meet the challenge of insufficient labeled samples in HSI classification tasks, we propose the ADSA-FSL framework, as shown in Fig. 1. Specifically, we first propose the ADSA, which is composed of an adversarial learning-based spectral encoder and an attribute-driven spectral decoder, to align different HSIs for obtaining numerous labeled samples with the same distribution as the TD HSI. Then, we design a 3-D RPN to model spectral–spatial structures of HSIs and combine it with an FSL framework to form the ADSA-FSL. In this section, we present them one by one in detail.

A. Adversarial Learning-Based Spectral Encoder

The ADSA allows for the input of M ($M \geq 1$) SD HSIs with one TD HSI simultaneously. To simplify the model description, we take the input of two HSIs (i.e., one SD HSI and one TD HSI) as an example. The dissimilar spectral attributes of heterogeneous HSIs, comprising wavelengths, dimensionality, and reflectance shifts, create hindrances in cross-domain transfer learning. To solve this issue, inspired by domain-adaptation techniques [37], [38], [39], we design the adversarial learning-based spectral encoder to embed different HSIs into a latent feature subspace and then reduce their distribution gap by adversarial learning.

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Let the spectra in SD $\mathcal{S}$ be $\mathcal{D}^{\mathcal{S}} = \{\mathbf{x}_i^{\mathcal{S}}\}$, where $\mathbf{x}_i^{\mathcal{S}} \in \mathbb{R}^{L^{\mathcal{S}}}$ represents the ith spectrum (in vector form) with a dimensionality of $L^{\mathcal{S}}$. Similarly, the spectra in TD $\mathcal{T}$ are given by $\mathcal{D}^{\mathcal{T}} = \{\mathbf{x}_i^{\mathcal{T}}\}$, where $\mathbf{x}_i^{\mathcal{T}} \in \mathbb{R}^{L^{\mathcal{T}}}$. Since the two datasets have different wavelengths and dimensionalities (i.e., $L^{\mathcal{S}} \neq L^{\mathcal{T}}$), two nonlinear transformations $f_{\text{encode}}^{\mathcal{S}} : \mathcal{S} \to \mathcal{F}$ and $f_{\text{encode}}^{\mathcal{T}} : \mathcal{T} \to \mathcal{F}$ are constructed to map them into the same feature space $\mathcal{F}$. In our model, $f_{\text{encode}}^{\mathcal{S}}$ and $f_{\text{encode}}^{\mathcal{T}}$ are represented as two independent multilayer perceptions (MLPs), as shown in Fig. 2(a). Then, the transformed spectral embeddings of SD and TD data can be expressed as follows:

$$\begin{aligned} \mathcal{D}_{\text{encode}}^{\mathcal{S}} &= \left\{\mathbf{e}_i^{\mathcal{S}} = f_{\text{encode}}^{\mathcal{S}}\left(\mathbf{x}_i^{\mathcal{S}}\right)\right\} \\ \mathcal{D}_{\text{encode}}^{\mathcal{T}} &= \left\{\mathbf{e}_i^{\mathcal{T}} = f_{\text{encode}}^{\mathcal{T}}\left(\mathbf{x}_i^{\mathcal{T}}\right)\right\} \end{aligned} \tag{1}$$

where $\mathbf{e}_i^{\mathcal{S}} \in \mathbb{R}^d$ and $\mathbf{e}_i^{\mathcal{T}} \in \mathbb{R}^d$ are the embeddings of the spectra $\mathbf{x}_i^{\mathcal{S}}$ and $\mathbf{x}_i^{\mathcal{T}}$, respectively, and d is the dimensionality of embeddings. Despite $\mathcal{D}_{\text{encode}}^{\mathcal{S}}$ and $\mathcal{D}_{\text{encode}}^{\mathcal{T}}$ having the same dimensionality, there still exists a significant distribution gap between them. To address this issue, we employ the concept of adversarial learning to further minimize the distribution distance between the two domains. Specifically, we first create a domain classifier f_{cls} composed of MLPs and a softmax layer [as shown in Fig. 2(b)], which aims to learn the distribution of the two domains by categorizing the domain label of each spectral embedding. This can be represented as follows:

$$\mathcal{L}_{\text{domain}} = \sum_i \left\| f_{\text{cls}}\left(\mathbf{e}_i^{\mathcal{S}}\right) - \mathbf{L}^{\mathcal{S}} \right\|_2 + \sum_i \left\| f_{\text{cls}}\left(\mathbf{e}_i^{\mathcal{T}}\right) - \mathbf{L}^{\mathcal{T}} \right\|_2 \tag{2}$$

where $\mathbf{L}^{\mathcal{S}}$ and $\mathbf{L}^{\mathcal{T}}$ represent the one-hot labels of the SD and TD embeddings, respectively, and $\mathcal{L}_{\text{domain}}$ is the domain classification loss. As there are only two domains (i.e., $\mathcal{S}$ and $\mathcal{T}$) in this example, we can set $\mathbf{L}^{\mathcal{S}} = [1, 0]^{\mathrm{T}}$, and $\mathbf{L}^{\mathcal{T}} = [0, 1]^{\mathrm{T}}$, respectively. Furthermore, we employ the technique of reverse gradient [41] to implement adversarial training, which can avoid the additional computation brought about by alternating training. Adversarial training can lead to spectral embeddings in the feature space $\mathcal{F}$ with similar distributions. To emphasize the reverse gradient training, we denote it as follows:

$$\mathcal{L}_{\text{domain}}^{-\text{grad}} = \begin{cases} \mathcal{L}_{\text{domain}}, & \text{in forward propagation} \\ -\mathcal{L}_{\text{domain}}, & \text{in backward propagation.} \end{cases} \tag{3}$$

By reversing the gradient of the domain classification loss in backpropagation, the representation of different HSIs in the feature space $\mathcal{F}$ can be confused, thus achieving a similar effect to that of adversarial learning.

B. Attribute-Driven Spectral Decoder

In [40], images acquired by different sensors are decoupled into sensor-independent content representations as well as sensor-related attribute representations, allowing the image representations to be controlled by adjusting the attributes. Inspired by this, we propose to decouple the domain-specific attributes of different HSIs, aiming to enable the network to control the spectral representations based on given attributes.

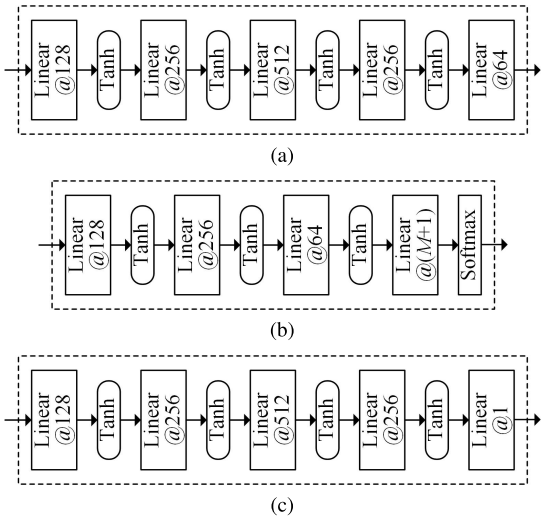


Fig. 2. Detailed network architectures of different nonlinear transformation modules in the ADSA. (a) Architecture of f_{encode}. (b) Architecture of f_{cls}, where $M = 1$ in this example. (c) Architecture of f_{mlp}.

Assuming that any spectral curve is an unfolding of its embedding under a given wavelength sequence, then we can regard the wavelength at each point of the spectrum as its position. In natural language processing, the position of a word in a sentence is crucial. To enable models to leverage the position information of each word, researchers have developed several predefined functions [42], [43] to embed the position information of words into networks. Inspired by this, we use the sinusoidal positional encoding [42] to add the wavelength information into spectral embeddings, thus enabling the network to learn the potential relationship between the wavelengths and the spectra. For the sake of description, the positional encoding of the spectrum is referred to as wavelength encoding in this article. Let the wavelength sequence of the SD dataset be $\lambda^{\mathcal{S}} = [\lambda_1^{\mathcal{S}}, \lambda_2^{\mathcal{S}}, \ldots, \lambda_{L^{\mathcal{S}}}^{\mathcal{S}}]^{\mathrm{T}}$, and its wavelength encoding $\mathbf{W}^{\mathcal{S}} \in \mathbb{R}^{L^{\mathcal{S}} \times d}$ can be calculated by

$$W_{j,p}^{\mathcal{S}} = \begin{cases} \sin\left(\lambda_j^{\mathcal{S}} / 5000^{2p/d}\right), & \text{if } p \text{ is even} \\ \cos\left(\lambda_j^{\mathcal{S}} / 5000^{2p/d}\right), & \text{if } p \text{ is odd.} \end{cases} \tag{4}$$

Then, the spectral embeddings $\mathcal{D}_{\text{encode}}^{\mathcal{S}}$ can be unfolded, in turn, with the wavelength encoding $\mathbf{W}^{\mathcal{S}}$ to reconstruct the original spectra $\mathcal{D}_{\text{decode}}^{\mathcal{S}} = \{\hat{x}_i^{\mathcal{S}}\}$. In addition, considering that there is still a gap in spectral reflectance shift between different HSIs due to factors, such as sensors and scenes, we assign a set of learnable parameters to correct each spectrum, i.e.,

$$\hat{x}_{i,j}^{\mathcal{S}} = k_j^{\mathcal{S}} \times f_{\text{mlp}}\left(\mathbf{e}_i^{\mathcal{S}} + \mathbf{W}_j^{\mathcal{S}}\right) + b_j^{\mathcal{S}} \tag{5}$$

where $\hat{x}_{i,j}^{\mathcal{S}}$ denotes the value of the reconstructed spectrum $\hat{x}_i^{\mathcal{S}}$ at the jth band, $k_j^{\mathcal{S}}$ and $b_j^{\mathcal{S}}$ are the scaling and offset values used to correct spectrum, and f_{mlp} is an MLP for nonlinear transformation shown in Fig. 2(c). Note that the reflectance shift coefficients ($\mathbf{K}^{\mathcal{S}}$ and $\mathbf{b}^{\mathcal{S}}$) are learnable parameters with initial values of all-1 vector and all-0 vector, respectively, and are updated with the network. For simple description, we notate the attributes, i.e., wavelength encoding and reflectance shift coefficients, of the SD HSI as

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$\mathcal{A}^{\mathcal{S}} = \{\mathbf{W}^{\mathcal{S}}, \mathbf{K}^{\mathcal{S}}, \mathbf{b}^{\mathcal{S}}\}$. Similarly, the attributes and reconstructed spectra of the TD HSI can be denoted as $\mathcal{A}^{\mathcal{T}} = \{\mathbf{W}^{\mathcal{T}}, \mathbf{K}^{\mathcal{T}}, \mathbf{b}^{\mathcal{T}}\}$ and $\mathcal{D}^{\mathcal{T}}_{\text{decode}} = \{\hat{x}_i^{\mathcal{T}}\}$, respectively. For the convenience of description, we record the process of encoding and decoding as follows:

$$\begin{aligned} \hat{x}_i^{\mathcal{S}} &= f_{\text{decode}}\left(f^{\mathcal{S}}_{\text{encode}}\left(\mathbf{x}_i^{\mathcal{S}}\right); \mathcal{A}^{\mathcal{S}}\right) \\ \hat{x}_i^{\mathcal{T}} &= f_{\text{decode}}\left(f^{\mathcal{T}}_{\text{encode}}\left(\mathbf{x}_i^{\mathcal{T}}\right); \mathcal{A}^{\mathcal{T}}\right) \end{aligned} \tag{6}$$

where f_{decode} denotes the attribute-driven spectral decoder, i.e., (5). The wavelength encoding $\mathbf{W}^{\mathcal{S}}$ and $\mathbf{W}^{\mathcal{T}}$ are calculated from given real wavelengths, while the reflectance shift coefficients $(\mathbf{K}^{\mathcal{S}}, \mathbf{b}^{\mathcal{S}})$ and $(\mathbf{K}^{\mathcal{T}}, \mathbf{b}^{\mathcal{T}})$ need to be learned from data in a self-reconstruction manner, i.e.,

$$\begin{aligned} \mathcal{L}_{\text{recon}} &= \mathcal{L}^{\mathcal{S}}_{\text{recon}} + \mathcal{L}^{\mathcal{T}}_{\text{recon}} \\ &= \sum_i \left\| \hat{x}_i^{\mathcal{S}} - \mathbf{x}_i^{\mathcal{S}} \right\|_2 + \sum_i \left\| \hat{x}_i^{\mathcal{T}} - \mathbf{x}_i^{\mathcal{T}} \right\|_2. \end{aligned} \tag{7}$$

Considering that adjacent HSI bands have high similarity, a smoothing prior term is designed to further enhance the modeling of the relationship between wavelengths and spectra. Specifically, since the wavelength range of a common HSI is within [300, 2500] nm, we construct an arithmetic sequence ranging from 300 to 2500 with step size 1 as a predefined wavelength sequence, i.e., $\lambda^{\mathcal{P}} = [300, 301, 302, \ldots, 2500]^{\mathrm{T}}$. Then, the corresponding wavelength encoding $\mathbf{W}^{\mathcal{P}}$ can be calculated by (4), and a smoothing constraint is applied to the spectra, i.e.,

$$\mathcal{L}_{\text{smooth}} = \sum_j \sum_{\mathbf{e}} \left\| f_{\text{mlp}}\left(\mathbf{e} + \mathbf{W}^{\mathcal{P}}_j\right) - f_{\text{mlp}}\left(\mathbf{e} + \mathbf{W}^{\mathcal{P}}_{j+1}\right) \right\|_1 \tag{8}$$

where $\mathbf{e}$ enumerates in $\mathcal{D}^{\mathcal{S}}_{\text{encode}} \bigcup \mathcal{D}^{\mathcal{T}}_{\text{encode}}$ and $\|\cdot\|_1$ denotes the $L1$ norm. Finally, the total training loss of ADSA is

$$\mathcal{L}_{\text{ADSA}} = \mathcal{L}_{\text{recon}} + \alpha \mathcal{L}^{-\text{grad}}_{\text{domain}} + \beta \mathcal{L}_{\text{smooth}} \tag{9}$$

where α and β are empirical parameters, which are fixed to 1 and 0.1 in all experiments, respectively.

Sufficiently rich sample classes are essential for the learning of transferable meta-knowledge. By simply extending the ADSA, we can align more than two HSIs simultaneously, as shown in Fig. 1. In practical applications, we first train the network in an unsupervised manner according to (9) and then convert each HSI into a specific spectral space by adjusting its attributes. Since our goal is to classify the TD HSI, we only need to align multiple SD HSIs to the TD HSI, in turn, i.e.,

$$\hat{x}_i^{\mathcal{S}_m \to \mathcal{T}} = f_{\text{decode}}\left(f^{\mathcal{S}_m}_{\text{encode}}\left(\mathbf{x}_i^{\mathcal{S}_m}\right); \mathcal{A}^{\mathcal{T}}\right) \tag{10}$$

where $\mathcal{D}^{\mathcal{S}_m} = \{\mathbf{x}_i^{\mathcal{S}_m}\}$ denotes the mth $(1 \le m \le M)$ SD dataset, $f^{\mathcal{S}_m}_{\text{encode}}$ is its corresponding encoder, and $\mathcal{D}^{\mathcal{S}_m \to \mathcal{T}} = \{\hat{x}_i^{\mathcal{S}_m \to \mathcal{T}}\}$ denotes the transferred spectra from SD $\mathcal{S}_m$ into TD $\mathcal{T}$. Then, numerous labeled samples with the same distribution as TD HSI can be obtained by combining different aligned SD HSIs, i.e., $\mathcal{D}_{\text{source}} = \bigcup_m \mathcal{D}^{\mathcal{S}_m \to \mathcal{T}}$. Similarly, the TD dataset can be noted as $\mathcal{D}_{\text{target}} = \mathcal{D}^{\mathcal{T}}$.

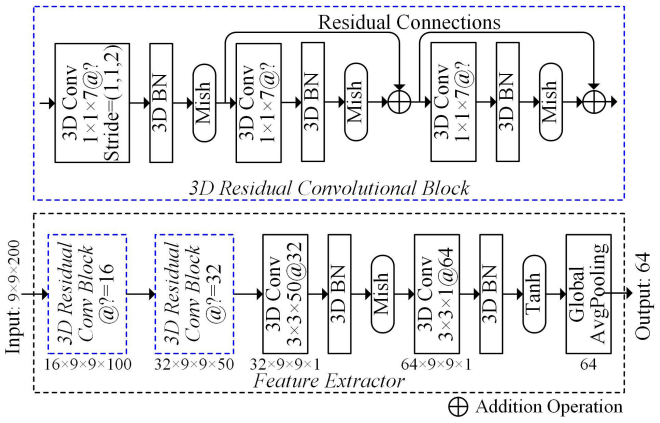


Fig. 3. Detailed network architecture of the feature extractor in 3-D RPN, i.e., f_{feature}, with the number of input bands equal to 200 for an example.

C. 3-D Residual Prototypical Network

Compared with the high-dimensional adaptive features, the local structure features of spectra have better generalizability [37], [38], [39]. Therefore, we build the 3-D RPN by utilizing the 3-D residual network [19] and the prototypical network [25]. The 3-D RPN consists of a feature extractor and a metric classifier, in which the former is used to model the geometric structures of HSIs and extract discriminative knowledge, and the latter can dynamically adjust the number of classes and enhance the small sample learning ability.

As shown in Fig. 3, in the feature extractor, we construct two 3-D residual convolutional blocks with 3-D convolutions and residual connections for the extraction of spectral–spatial structure features, followed by several convolutional layers and a global average pooling (AvgPooling) layer to embed samples into a feature subspace. The 3-D batch normalization (BN) is used to enhance the antioverfitting capability of the network. The Mish [44] is used as the activation function in each convolutional layer. Before the global AvgPooling layer, we use the tanh function to compress these embeddings to $[-1, 1]$ to improve the training stability. Since all the operations used in the network are regular, we do not describe them in detail here. For ease of description, we denote the feature extractor as f_{feature}.

The metric classifier in the prototypical network is a common method for small sample classification tasks, which can improve feature generalization and classification robustness by optimizing the discriminative feature distribution and interclass decision boundaries. Compared with the softmax classifier, the metric classifier can directly predicate the class labels based on the distance between samples and the centroid of each class, without learning a linear transformation layer. Therefore, it is suitable for classification tasks with variable classes. In FSL, samples are used to train the network in the form of episodes. Specifically, a C-way K-shot episode consists of a support set and a query set, each of which contains C classes, and each class contains K samples. Let the support set in an episode be $\mathcal{E}^{\text{spt}} = \{\mathbf{X}^{\text{spt}}_{c,k} | 1 \le c \le C, 1 \le k \le K\}$, where $\mathbf{X}^{\text{spt}}_{c,k} \in \mathbb{R}^{H \times W \times B}$ is the cth class kth sample in cube form (distinct from the spectral-form samples in ADSA), and H, W, and B denote the height, width, and bands of

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Algorithm 1 Procedure of ADSA-FSL

Input: SD HSIs $\mathcal{D}^{\mathcal{S}_1}, \ldots, \mathcal{D}^{\mathcal{S}_M}$ and TD HSI $\mathcal{D}^{\mathcal{T}}$
Output: Classification results of TD HSI
// Train ADSA with unlabeled SD and TD samples
for *Iteration* = *1* $\rightarrow$ *Max_Iter_ADSA* **do**
  Randomly select a batch of unlabelled spectra from $\mathcal{D}^{\mathcal{S}_1}, \ldots, \mathcal{D}^{\mathcal{S}_M}, \mathcal{D}^{\mathcal{T}}$ to minimize loss $\mathcal{L}_{ADSA}$
// Align each SD HSI to TD HSI
for $m = 1 \rightarrow M$ **do**
  Generate $\mathcal{D}^{\mathcal{S}_m \rightarrow \mathcal{T}}$ according to Eq. (10)
$\mathcal{D}_{source} = \bigcup_m \mathcal{D}^{\mathcal{S}_m \rightarrow \mathcal{T}}$, $\mathcal{D}_{target} = \mathcal{D}^{\mathcal{T}}$
// Train 3D-RPN with labeled SD and TD samples
for *Iteration* = *1* $\rightarrow$ *Max_Iter_FSL* **do**
  Randomly select an episode from $\mathcal{D}_{source}$ to minimize loss $\mathcal{L}_{FSL}$
  Randomly select an episode from $\mathcal{D}_{target}$ to minimize loss $\mathcal{L}_{FSL}$
// Classify TD HSI
Compute prototypes based on labeled samples in $\mathcal{D}_{target}$ according to Eq. (11)
Predict labels of testing samples in $\mathcal{D}_{target}$ according to Eq. (12)

each 3-D cube sample. Similarly, the query set is denoted as $\mathcal{E}^{\text{que}} = \{\mathbf{X}_{c,k}^{\text{que}}\}$, which has the same classes and number of samples as $\mathcal{E}^{\text{spt}}$. The prototype of each class can be computed by averaging the sample embeddings of this class, i.e.,

$$\mathbf{G}_c = \frac{1}{K} \sum_k f_{\text{feature}}\left(\mathbf{X}_{c,k}^{\text{spt}}\right) \tag{11}$$

where $\mathbf{G}_c$ denotes the prototype of the cth class. Subsequently, the query samples can be classified based on the metric distances to each prototype. In contrast to the use of Euclidean distance as the metric distance [37], [38], we consider the prototypes as bases in the feature space and then predict the classes to which the samples belong by comparing the projection length of their features on each base, i.e.,

$$S_{c,k} = f_{\text{feature}}\left(\mathbf{X}_{c,k}^{\text{que}}\right) \cdot \frac{\mathbf{G}_c}{\|\mathbf{G}_c\|_2} \tag{12}$$

where the operator $\cdot$ denotes inner product and $S_{c,k}$ is the projection length of the sample $\mathbf{X}_{c,k}^{\text{que}}$ on the cth prototypical base. Correspondingly, its probability is denoted as $P_{c,k} = \text{Softmax}(S_{c,k})$. Then, we can define the loss function as follows:

$$\mathcal{L}_{\text{FSL}} = -\sum_{c=1}^{C} \sum_{k=1}^{K} L_{c,k} \log\left(P_{c,k}\right) \tag{13}$$

where $L_{c,k} \in \{0, 1\}$ is the true probability that the query sample $\mathbf{X}_{c,k}^{\text{que}}$ belongs to the cth class. Once the network training is completed, the similarity of the testing samples with each prototype can be calculated by (12), as well as determining the class to which they belong.

D. FSL and Classification

As shown in Fig. 1, the proposed ADSA method is able to align the spectral attributes of different HSIs, so that the samples in different datasets are in the same distribution while preserving their inherent structures. Subsequently, we can utilize those aligned labeled SD samples to assist in improving the classification performance of TD HSI within an FSL framework.

Our method uses the same training approach as [37], [38], and [39]. For a C-way K-shot task, it aims to learn a C-class classifier using K labeled samples per class. In the small-sample classification of HSI, C denotes the number of the TD classes, and K is usually set to a small value, e.g., 1 or 5. In the training process, C classes are randomly selected from the SD dataset $\mathcal{D}_{\text{source}}$ each time, with $2K$ samples per class, as an episode, which are then divided equally into two parts with K samples per class as the support set and the query set, respectively. Meanwhile, for the TD dataset $\mathcal{D}_{\text{target}}$, since only K labeled samples per class are available, we use a rotation strategy to augment the samples to meet the training requirements. The episodes collected from the SD and TD dataset are, in turn, fed to 3-D RPN to train the network. After training, we can calculate the prototype of each TD class based on the labeled samples in $\mathcal{D}_{\text{target}}$ and classify the remaining test samples accordingly. The overall FSL and classification procedure is described in Algorithm 1.

IV. Experiments

To evaluate the proposed method, we used four public datasets as benchmarks, which are Indian Pines, Salinas, Botswana, and Chikusei. Also, some representative methods as well as recent advanced methods are selected as comparison methods. The classification network used in ADSA-FSL is designed based on the prototypical network, and if the feature extractor is removed, it will degenerate into a K-nearest neighbors (KNN)-like method. The 2-D CNN [16] and 3-D CNN [17] use different convolutional architectures. Spectral-spatial residual network (SSRN) [19] is constructed by further introducing residual connections on top of regular CNNs. Deep few-shot learning (DFSL) [28] is a method constructed based on a prototypical network for HSI small-sample classification. The DCFSL [37], GiaCFSL [38], and CACFSL [39] are recent advanced methods for HSI cross-domain FSL and classification. We adopt the overall accuracy (OA), average accuracy (AA), and kappa statistic (KPP) as metrics to evaluate the performance of each method.

A. Experimental Configuration

In the training stage of ADSA, we randomly select 10 000 unlabeled spectra from each HSI to train the network using the RMSprop optimizer with a learning rate equal to $1e-5$, iterations equal to 100, and batch size equal to 64. For the reflectance shift coefficients, we set their learning rate to 0.0025 for accelerated training. In the training stage of FSL, K is set to 5, and C is set to be the number of categories of the TD dataset, and the patch size of each sample is fixed to 9×9 (i.e., $H = W = 9$). Besides, the Adam optimizer is used to optimize the 3-D RPN with a learning rate equal to $1e-3$ and

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TABLE I
CATEGORY INFORMATION OF DIFFERENT DATASETS

Class No.	Indian Pines		Salinas		Botswana		Chikusei	
	Name	Samples	Name	Samples	Name	Samples	Name	Samples
1	Alfalfa	54	Brocoli_green_weeds_1	2009	Water	270	Water	722
2	Corn-notill	1434	Brocoli_green_weeds_2	3726	Hippo grass	101	Bare soil (school)	714
3	Corn-mintill	834	Fallow	1976	Floodplain grasses1	251	Bare soil (park)	73
4	Corn	234	Fallow_rough_plow	1394	Floodplain grasses2	215	Bare soil (farmland)	1212
5	Pasture	497	Fallow_smooth	2678	Reeds1	269	Natural plants	1074
6	Trees/Grass	747	Stubble	3959	Riparian	269	Weeds in farmland	276
7	Pasture-mowed	26	Celery	3579	Fierscar2	259	Forest	5132
8	Hay-windrowed	489	Grapes_untrained	11271	Island interior	203	Grass	1642
9	Oats	20	Soil_vinyard_develop	6203	Acacia woodlands	314	Rice field (grown)	3340
10	Soybeans-notill	968	Corn_senesced_green_weeds	3278	Acacia shrublands	248	Rice field (first stage)	315
11	Soybean-mintill	2468	Lettuce_romaine_4wk	1068	Acacia grasslands	305	Row crops	1487
12	Soybean-cleantill	614	Lettuce_romaine_5wk	1927	Short mopane	181	Plastic house	549
13	Wheat	212	Lettuce_romaine_6wk	916	Mixed mopane	268	Manmade (non-dark)	302
14	Woods	1294	Lettuce_romaine_7wk	1070	Exposed soils	95	Manmade (dark)	1917
15	Building-Grass	380	Vinyard_untrained	7268			Manmade (blue)	112
16	Stone-steelTowers	95	Vinyard_vertical_trellis	1807			Manmade (red)	57
17							Manmade grass	258
18							Asphalt	203
19							Paved ground	37
Total	16 classes with 10366 samples		16 classes with 54129 samples		14 classes with 3248 samples		19 classes with 19422 samples	

iterations equal to 2000. Accordingly, 2000 episodes were randomly sampled from SD HSIs and TD HSI, respectively, for training, i.e., $N = 4000$. For each TD HSI (e.g., Indian Pines), the remaining three HSIs are used as SD datasets (e.g., Salinas, Botswana, and Chikusei). Our code is implemented using PyTorch 1.9 and Python 3.9, running on the platform of I7-8700K CPU with 32-GB memory and GTX-1080Ti GPU with 11-GB video memory.

B. Datasets

We adopt four public HSIs as the benchmarks, they are as follows.

1) *Indian Pines*[1]*:* This dataset is composed of 200 imagery bands with a wavelength range of 400–2490 nm and an image size of 145 × 145 pixel, which was captured by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) sensor over the agricultural site in the Indian Pines. There are 10 366 labeled samples of 16 terrain classes contained in this HSI.
2) *Salinas* [1]*:* This dataset contains 204 bands with a wavelength range of 390–2490 nm and an image size of 512 × 217 pixel, which was captured by the AVIRIS sensor over the Salinas Valley. A total of 54 129 samples of 16 crops are labeled for the research.
3) *Botswana* [1]*:* This dataset is composed of 145 imagery bands with a wavelength range of 447–2356 nm and an image size of 1476 × 256 pixel, which was obtained by the Hyperion sensor on the NASA EO-1 satellite over the Okavango Delta, Botswana. There are 3248 labeled samples of 14 classes contained in this HSI.
4) *Chikusei*[2]*:* This dataset includes 128 bands with a wavelength range of 362–1018 nm and an image size of 2517 × 2335 pixel, which was captured by the Headwall Hyperspec-VNIR-CIRIS imaging sensor over Chikushi, Japan. To facilitate the experiments, we reduced its height and width to 1/2 of its original size (i.e., 1259 × 1168); thus, there are 19 422 samples of 19 terrain classes retained in this dataset.

Note that for the DCFSL, GiaCFSL, and CACFSL, we use the raw Chikusei dataset as the SD dataset of Indian Pines, Salinas, and Botswana according to the settings in their original papers. In addition, in the experiments on Chikusei, since both methods require the number of classes in the SD to be larger than the TD, we use the XiongAn[3] dataset with 20 classes as the SD dataset, which was acquired by the full-spectrum multimodal imaging spectrometer over Matiwan Village, Xiong'an New Area, China. Detailed category information of other four datasets is displayed in Table I.

C. Comparison on Classification Performance

We test each method on four public datasets, in which each experiment is repeated ten times to report the mean and standard deviation of each index, as shown in Tables II–V. Among them, the indices in bold denote the best, and underlined ones are the second best. Correspondingly, the classification maps of different methods on each HSIs are shown in Figs. 4–7.

1) Indian Pines: In the experiments on this dataset, our method is significantly superior to other methods. This dataset has high interclass similarity and intraclass spectral variations, needing to fully exploit its discriminative features. The KNN demonstrates the results of directly classifying it, while 2-D CNN and 3-D CNN do not achieve a significant lead in comparison, indicating that the deep network is susceptible to overfitting under small samples, whereas SSRN can alleviate this problem due to the residual architecture. The DFSL constructed based on the prototypical network also performs better due to the usage of metric learning. However, the above methods are limited in performance, because they can only

[1] www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes
[2] www.naotoyokoya.com/Download.html
[3] http://www.hrs-cas.com/a/share/shujuchanpin/2019/0501/1049.html

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TABLE II
INDIVIDUAL CLASS, OA, AA, AND KPP OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class	KNN	2D-CNN [16]	3D-CNN [17]	SSRN [19]	DFSL [28]	DCFSL [37]	GiaCFSL [38]	CACFSL [39]	Ours
1	78.37±9.18	28.50±11.1	30.73±12.5	50.20±18.5	82.45±11.5	89.59±13.8	89.80±11.9	93.47±5.69	**99.80±0.61**
2	29.31±7.49	47.27±14.5	37.67±13.8	61.50±20.7	31.49±11.5	46.98±11.9	48.16±12.5	52.55±11.5	**61.64±15.2**
3	27.82±8.82	29.64±4.86	24.06±3.41	49.98±14.4	33.26±11.4	51.66±11.3	46.49±12.8	60.34±12.3	**60.82±12.8**
4	32.18±5.33	22.88±9.22	15.52±3.07	35.09±11.5	48.47±13.5	71.05±12.7	70.52±17.5	82.12±8.74	**95.28±6.89**
5	55.41±18.2	49.80±11.1	44.26±8.66	**79.52±13.4**	58.68±16.6	65.69±15.6	67.40±15.5	74.85±7.63	76.14±9.32
6	70.54±13.4	77.80±9.89	70.92±10.9	93.09±5.53	81.81±8.84	81.95±8.75	78.77±8.72	91.81±2.52	**93.13±3.13**
7	96.19±5.95	20.53±8.83	17.60±8.93	31.37±17.4	96.67±4.29	98.57±3.05	99.05±2.86	**100±0**	**100±0**
8	86.40±11.6	93.72±4.89	90.76±4.44	**99.28±1.08**	85.83±6.81	84.86±8.39	80.52±8.15	89.21±6.08	97.91±4.52
9	98.00±6.00	18.04±14.2	10.91±2.53	23.02±15.8	94.67±9.33	**100±0**	98.67±2.67	97.33±5.33	98.67±4.00
10	47.83±10.3	38.08±9.31	30.10±9.72	58.65±17.3	50.81±9.59	60.16±16.2	67.26±13.5	68.74±14.7	**72.51±11.5**
11	51.13±14.2	60.63±4.21	52.70±6.21	**70.84±9.86**	50.51±9.11	53.03±5.32	59.77±5.50	58.37±9.22	66.18±8.86
12	23.10±6.19	26.54±7.51	24.58±3.86	50.05±10.6	32.00±14.9	38.51±11.5	41.59±13.3	52.99±6.73	**67.26±9.85**
13	85.99±10.5	65.03±18.3	49.69±13.7	93.52±5.45	95.07±5.06	94.01±10.3	97.00±3.96	95.07±8.66	**99.28±1.19**
14	55.62±18.1	81.53±8.10	81.36±4.70	**94.79±3.61**	75.97±10.9	84.40±14.1	80.61±14.9	90.40±4.34	91.06±5.75
15	26.69±10.3	46.87±6.41	36.10±8.64	61.35±16.1	50.96±6.18	66.11±17.8	71.09±11.7	78.72±18.8	**91.89±5.16**
16	74.44±8.46	59.92±18.3	39.67±15.7	77.67±20.8	98.56±1.49	98.11±2.17	99.22±1.32	99.12±6.00	**99.89±0.33**
OA(%)	47.96±4.22	50.46±3.76	43.48±3.70	62.92±2.95	54.51±3.93	62.44±4.11	64.06±4.99	69.71±3.34	**75.63±2.97**
AA(%)	58.69±2.34	47.92±3.30	41.04±3.22	64.37±4.14	66.70±3.83	74.04±2.72	74.74±3.53	80.32±2.46	**85.72±1.49**
KPP(×100)	41.34±4.18	44.25±3.98	37.03±3.75	58.73±2.93	49.00±4.29	57.80±4.67	59.52±5.62	66.42±3.52	**72.62±3.27**

TABLE III
INDIVIDUAL CLASS, OA, AA, AND KPP OF DIFFERENT METHODS ON THE SALINAS DATASET

Class	KNN	2D-CNN [16]	3D-CNN [17]	SSRN [19]	DFSL [28]	DCFSL [37]	GiaCFSL [38]	CACFSL [39]	Ours
1	85.88±5.47	70.10±24.6	85.03±16.7	92.82±18.2	97.62±5.14	99.04±0.99	98.54±2.15	99.48±0.70	**99.55±0.78**
2	95.91±2.35	75.20±19.3	93.09±7.91	**99.98±0.06**	99.65±0.50	99.68±0.36	99.62±0.37	99.78±0.13	**99.98±0.04**
3	88.87±11.0	92.09±2.91	85.83±6.54	93.57±5.40	97.48±5.01	90.13±10.7	87.93±9.46	94.86±4.30	**98.04±4.91**
4	96.00±2.58	94.81±2.93	71.79±11.9	95.64±3.46	98.59±1.97	**99.64±0.37**	99.09±0.82	99.61±0.49	98.63±2.34
5	84.61±7.14	94.56±4.94	95.01±5.09	95.66±7.21	90.42±7.14	90.97±3.58	90.53±3.67	90.68±2.24	**95.72±3.04**
6	98.57±1.80	99.36±0.89	98.39±1.92	**99.93±0.10**	99.77±0.60	99.35±0.55	98.32±1.80	99.32±0.94	99.86±0.36
7	94.49±2.30	97.23±2.83	94.22±4.25	99.42±0.78	99.31±0.62	98.16±2.88	99.14±0.84	99.49±0.31	**99.76±0.13**
8	54.86±16.4	66.19±10.9	65.11±5.16	**84.54±9.74**	58.13±14.6	74.33±10.6	74.36±10.8	79.24±4.14	74.57±11.4
9	96.36±1.65	96.41±2.10	94.29±2.34	99.10±0.61	**99.39±0.79**	99.13±1.15	98.06±2.42	99.31±1.16	99.33±0.40
10	70.26±9.86	88.31±5.27	83.61±6.72	**94.07±2.58**	89.57±6.91	85.64±5.94	82.69±9.37	87.25±3.64	88.22±6.14
11	80.87±10.1	70.04±7.47	57.60±16.9	92.15±2.72	96.50±4.22	98.47±1.42	97.39±1.32	98.98±0.48	**99.47±0.57**
12	97.72±2.55	89.74±7.49	91.96±6.97	97.93±1.63	97.34±5.03	**99.76±0.37**	98.07±3.40	99.56±0.80	99.10±0.69
13	**99.56±0.39**	88.53±16.7	94.09±7.18	93.01±3.92	97.26±3.03	99.26±0.62	97.07±3.88	98.88±0.76	98.84±1.67
14	95.13±1.16	93.41±4.40	68.11±11.2	96.60±4.81	97.15±2.54	**99.10±0.76**	97.77±1.79	98.46±0.86	98.98±0.97
15	64.55±12.9	43.73±5.64	46.80±5.79	57.66±9.60	65.60±12.2	75.79±8.12	75.47±6.31	69.02±9.42	**86.91±7.12**
16	83.54±3.20	85.41±12.5	90.89±9.77	**99.00±1.90**	94.94±3.15	89.22±6.15	88.52±8.44	93.31±5.49	97.54±2.73
OA(%)	79.95±2.09	76.02±2.58	77.34±2.89	86.22±2.36	84.73±2.81	88.96±2.01	88.30±2.30	89.57±1.26	**91.62±2.51**
AA(%)	86.70±1.12	84.07±3.71	82.24±2.62	93.19±1.33	92.42±1.27	93.60±1.11	92.66±1.64	94.20±0.88	**95.91±1.09**
KPP(×100)	77.80±2.24	73.52±2.88	74.90±3.16	84.75±2.60	83.07±3.09	87.73±2.20	87.02±2.53	88.40±1.41	**90.71±2.76**

TABLE IV
INDIVIDUAL CLASS, OA, AA, AND KPP OF DIFFERENT METHODS ON THE BOTSWANA DATASET

Class	KNN	2D-CNN [16]	3D-CNN [17]	SSRN [19]	DFSL [28]	DCFSL [37]	GiaCFSL [38]	CACFSL [39]	Ours
1	97.21±1.88	95.40±5.20	86.66±8.02	99.25±0.60	**100±0**	99.96±0.11	99.89±0.34	**100±0**	99.70±0.47
2	98.02±2.61	65.60±17.4	54.38±22.3	85.42±22.2	97.29±3.49	99.48±1.07	99.06±1.43	**100±0**	98.85±3.44
3	96.22±4.45	95.93±4.51	91.60±4.22	**98.45±4.01**	94.96±4.96	98.41±2.17	97.89±1.62	96.75±3.12	97.52±5.78
4	97.19±2.34	89.96±5.33	82.29±18.7	92.79±4.99	97.10±4.22	99.48±0.50	98.62±1.47	99.24±1.30	**99.71±0.61**
5	42.65±9.14	93.98±5.56	74.51±8.31	93.17±5.06	69.77±13.3	80.76±8.34	86.17±7.56	82.12±8.33	**94.17±3.88**
6	47.88±8.32	75.01±7.47	55.80±16.1	89.89±6.76	83.64±13.2	93.90±8.46	**94.85±7.89**	93.26±11.4	94.77±4.48
7	81.93±10.2	97.26±3.49	97.20±4.31	**100±0**	96.06±5.53	99.72±0.40	99.72±0.61	99.61±0.35	99.80±0.47
8	76.67±8.19	95.59±4.89	92.21±8.52	96.58±8.99	95.56±7.24	94.90±9.85	92.53±10.9	**100±0**	97.12±8.64
9	38.19±9.55	88.04±8.92	63.23±13.2	92.88±9.29	87.90±9.36	94.72±7.40	93.79±4.98	93.53±8.89	**99.61±0.64**
10	96.17±5.06	94.66±4.59	73.67±18.7	91.73±8.31	92.43±7.91	99.92±0.16	99.71±0.58	**100±0**	99.34±1.84
11	91.30±4.72	91.49±7.68	79.07±8.59	98.88±2.41	96.80±1.21	**99.43±0.65**	98.57±2.73	99.33±0.56	99.23±1.42
12	87.95±8.45	79.29±8.53	60.85±11.2	98.79±1.64	98.41±1.24	98.98±1.62	98.69±1.42	97.84±2.34	**99.38±1.69**
13	48.02±16.3	98.09±2.64	79.92±13.9	95.13±6.70	99.13±1.20	98.63±1.68	97.98±2.50	99.70±0.61	**100±0**
14	85.56±7.32	94.50±8.31	76.11±20.4	97.22±6.79	95.56±4.82	93.56±7.08	92.89±6.85	90.44±11.2	**93.89±6.93**
OA(%)	74.74±1.30	88.93±2.74	75.17±8.82	94.34±2.71	92.53±2.80	96.45±1.01	96.44±1.10	96.51±1.01	**98.25±0.75**
AA(%)	77.50±1.25	89.63±2.38	76.25±8.79	95.01±2.58	93.18±2.51	96.56±1.13	96.45±1.07	96.56±0.65	**98.08±0.93**
KPP(×100)	72.71±1.40	88.02±2.96	73.15±9.53	93.87±2.93	91.91±3.03	96.15±1.09	96.14±1.19	96.22±1.09	**98.10±0.81**

learn classification patterns from this single HSI. In contrast, meta-learning-based DCFSL, GiaCFSL, and CACFSL can acquire prior knowledge from the Chikusei dataset, which effectively improves the classification accuracies. Also, our

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TABLE V
INDIVIDUAL CLASS, OA, AA, AND KPP OF DIFFERENT METHODS ON THE CHIKUSEI DATASET

Class	KNN	2D-CNN [16]	3D-CNN [17]	SSRN [19]	DFSL [28]	DCFSL [37]	GiaCFSL [38]	CACFSL [39]	Ours
1	45.10±10.2	64.42±16.1	70.20±10.6	**94.52±7.80**	84.66±8.85	76.62±13.7	81.02±12.8	82.71±8.10	93.01±11.7
2	77.21±17.7	73.03±12.3	69.90±15.0	97.31±5.65	97.55±2.33	92.05±7.25	91.44±5.51	95.74±1.65	**99.13±1.93**
3	70.59±10.4	19.16±12.0	10.46±4.09	23.17±11.9	**96.47±3.55**	82.35±11.7	87.65±9.80	89.12±5.93	91.47±16.3
4	74.86±14.1	67.57±9.85	59.51±11.9	**82.02±9.01**	61.63±11.0	74.23±11.9	69.22±14.9	77.88±8.05	71.14±12.6
5	88.36±6.60	89.46±4.82	80.42±9.56	94.94±4.84	99.67±0.46	98.35±1.58	96.10±4.09	98.88±0.58	**99.85±0.45**
6	90.63±3.81	60.02±22.2	42.78±18.5	70.73±32.1	92.80±2.47	91.33±4.97	92.25±3.14	92.25±4.06	**93.47±3.11**
7	43.87±19.0	90.00±5.98	84.47±9.70	94.89±6.27	96.13±3.51	99.10±0.61	99.04±1.20	**99.79±0.21**	99.41±1.02
8	61.66±11.3	89.02±9.21	91.55±5.28	**99.71±0.31**	89.32±4.37	91.03±3.21	86.26±4.97	89.98±1.27	94.43±2.19
9	88.01±10.8	**96.07±5.11**	59.53±14.0	94.81±8.80	82.90±15.7	94.29±5.35	90.85±7.79	94.49±5.47	93.59±6.24
10	78.58±10.8	64.69±17.1	31.30±8.05	78.85±12.0	98.16±3.76	90.90±6.73	92.81±7.77	95.94±5.10	**99.81±0.41**
11	67.52±10.6	**93.70±4.30**	78.06±6.98	93.31±6.42	62.38±10.7	73.91±14.8	72.76±10.5	80.11±8.53	78.40±11.2
12	33.71±8.96	85.69±7.97	45.39±13.2	**94.82±6.12**	57.28±8.99	72.56±7.71	69.03±12.2	75.22±9.49	78.29±11.2
13	74.21±10.6	69.13±10.2	66.71±11.4	94.34±9.46	92.76±3.37	93.54±2.15	90.94±4.34	95.29±0.30	**95.96±1.38**
14	82.79±12.0	94.73±4.61	95.80±2.25	**99.84±0.32**	93.03±5.56	88.25±5.79	87.40±6.52	93.19±3.57	96.46±3.46
15	91.50±5.04	59.01±15.4	63.30±18.0	78.97±17.7	98.22±2.56	94.86±6.83	95.42±6.68	95.51±4.36	**99.72±0.84**
16	69.42±9.00	83.67±12.5	83.23±18.0	82.75±27.6	89.81±11.3	88.46±10.9	87.31±11.7	86.54±11.0	**91.92±11.9**
17	35.34±9.35	82.52±11.6	30.80±12.4	93.28±13.3	92.33±6.44	88.34±6.68	87.83±6.97	95.02±4.08	**99.64±1.07**
18	64.24±9.23	65.62±10.8	41.27±11.2	70.62±13.6	83.89±9.87	80.10±13.7	81.41±14.8	80.10±13.2	**84.04±16.0**
19	45.31±14.2	29.31±10.6	8.75±4.93	41.02±35.3	95.31±5.63	94.69±4.65	96.88±4.64	96.25±7.50	**100±0**
OA(%)	66.24±5.37	83.11±2.84	66.43±5.16	89.72±3.50	86.70±3.23	90.37±1.91	88.82±2.20	92.39±0.64	**93.17±1.89**
AA(%)	67.52±2.70	72.46±2.98	58.60±3.51	83.15±3.83	87.60±1.81	87.63±2.20	87.14±2.16	90.21±1.55	**92.62±2.56**
KPP(×100)	61.71±5.89	80.50±3.30	61.95±5.86	88.22±3.92	84.74±3.67	88.90±2.19	87.13±2.53	91.22±0.73	**92.14±2.16**

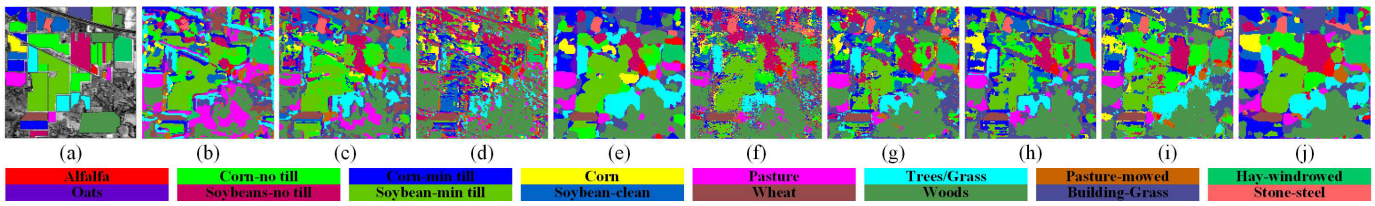


Fig. 4. Classification maps of different methods on the Indian Pines dataset. (a) Ground truth. (b) KNN. (c) 2-D CNN. (d) 3-D CNN. (e) SSRN. (f) DFSL. (g) DCFSL. (h) GiaCFSL. (i) CACFSL. (j) Ours.

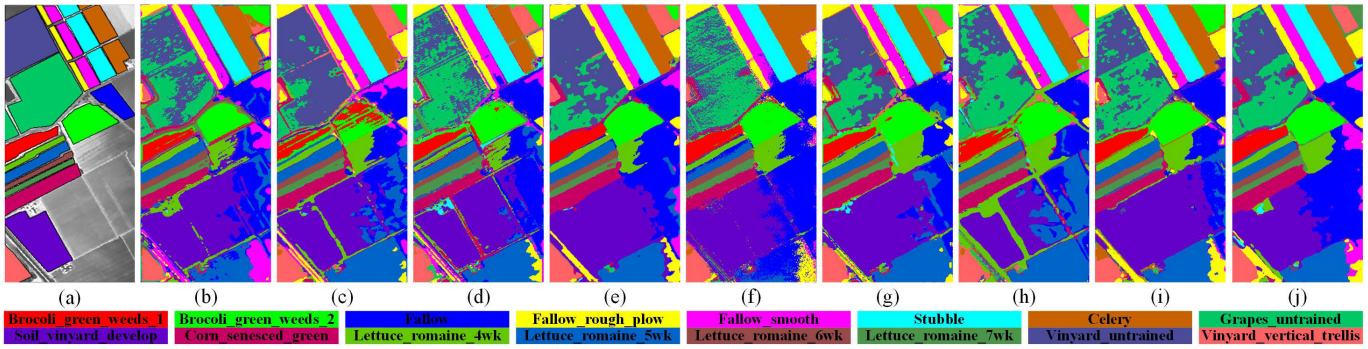


Fig. 5. Classification maps of different methods on the Salinas dataset. (a) Ground truth. (b) KNN. (c) 2-D CNN. (d) 3-D CNN. (e) SSRN. (f) DFSL. (g) DCFSL. (h) GiaCFSL. (i) CACFSL. (j) Ours.

method can acquire knowledge from multiple SD HSIs simultaneously, thereby further improving the performance.

2) Salinas: Our method also achieves leading results on this dataset. The boundaries of land covers in Salinas are clear and piecewise smoothed, so the performance of 2-D CNN and 3-D CNN on this dataset is significantly improved compared with that on the Indian Pines. Because of the residual connections, SSRN achieves the optimal classification performance on a single HSI. Based on this, by further introducing the prior knowledge from SD dataset, the performance of DCFSL, GiaCFSL, and CACFSL is further improved. However, the acquisition of knowledge from a single SD HSI is not sufficient, resulting in their accuracies still not outperforming our method. In contrast, our method can obtain results with better consistency with the ground truth.

3) Botswana: On this dataset, our method also shows great performance. This dataset has a complex and varied topography, and some of its rare classes pose a challenge to the classification model. However, by introducing the help of multiple-domain knowledge, our method successfully meets this challenge. As shown in Table IV, the performance of 2-D CNN and 3-D CNN as well as KNN is relatively low on

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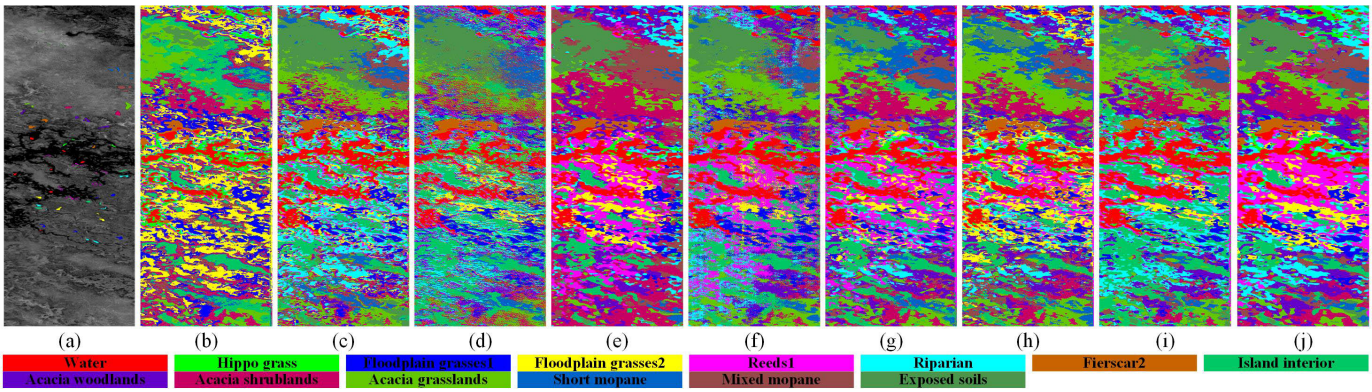


Fig. 6. Classification maps of different methods on the Botswana dataset. (a) Ground truth. (b) KNN. (c) 2-D CNN. (d) 3-D CNN. (e) SSRN. (f) DFSL. (g) DCFSL. (h) GiaCFSL. (i) CACFSL. (j) Ours.

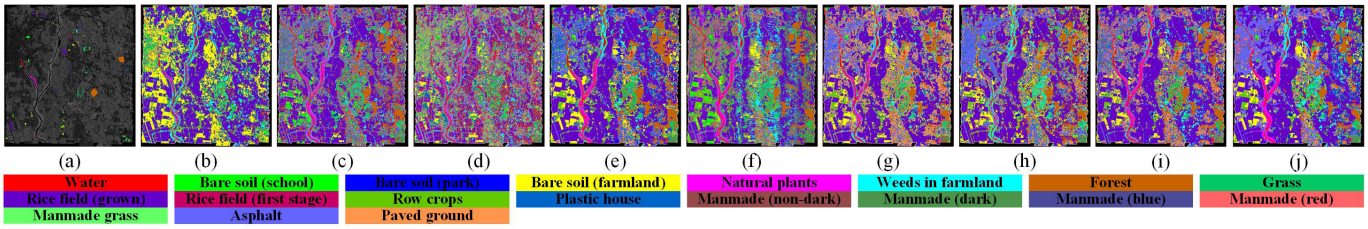


Fig. 7. Classification maps of different methods on the Chikusei dataset. (a) Ground truth. (b) KNN. (c) 2-D CNN. (d) 3-D CNN. (e) SSRN. (f) DFSL. (g) DCFSL. (h) GiaCFSL. (i) CACFSL. (j) Ours.

this dataset, whereas the performance of SSRN is significantly improved due to the residual architecture. DCFSL, GiaCFSL, and CACFSL are improved by utilizing the external prior knowledge, but their results still fail to reach the performance of our method. This result proves that learning from only a single source HSI is not sufficient; by contrast, learning from multiple SD datasets simultaneously is needed to achieve better discriminative knowledge.

4) Chikusei: On this dataset, our method continues to show leading performance. This dataset is characterized by a large number of samples and a wide distribution, which poses further difficulties for deep network training. As shown in Table V, due to the model requirement, we experimentally let DCFSL, GiaCFSL, and CACFSL acquire knowledge from the XiongAn dataset with 20 classes of land-cover samples. Since the sample classes of the SD HSI (i.e., XiongAn) do not greatly richer than those of the TD HSI (i.e., Chikusei), these three methods do not achieve superior results compared with ours. Because of ADSA, our method can obtain more diversified knowledge from SD HSIs with a much larger number of categories than the TD HSI and, thus, has better generalization performance, which further confirms the effectiveness of our method.

V. Discussions

In this section, we will further reveal and analyze the effectiveness of ADSA-FSL from several perspectives, including ablation experiments, visualization of feature distributions, analysis of feature transferability, generalizability testing of ADSA, analysis of hyperparameters, robustness to noise, and running efficiency.

A. Ablation Experiments

Different from the methods using a single SD HSI, our method can learn rich discriminative knowledge from multiple SD HSIs to further enhance classification performance. To verify the ability of the proposed method to obtain knowledge from SD HSIs, we conduct a set of ablation experiments. Specifically, we use a prototypical network with the same architecture as 3-D RPN as the baseline and compare its performance with ADSA-FSL using one SD HSI as well as that using all SD HSIs. For Indian Pines, Salinas, and Botswana, with reference to DCFSL, GiaCFSL, and CACFSL, we use Chikusei as the single SD dataset. For Chikusei, we use XiongAn as the single SD dataset.

As shown in Table VI, using one SD dataset can already effectively improve the accuracy of the model compared with baseline, which is also verified on DCFLS, GiaCFSL, and CACFSL. Specifically, with one SD dataset, the classification model improves the OA indices on Indian Pines, Salinas, Botswana, and Chikusei by 3.86%, 3.24%, 0.85%, and 0.39%, respectively, compared with the baselines. On this basis, when multiple SD datasets are used, the classification model further achieves 9.59%, 5.41%, 2.81%, and 3.36% improvement over baseline on the four datasets. These results show that our method can effectively improve the performance by learning prior knowledge from SD datasets. Moreover, in contrast to the methods of learning knowledge from a single SD HSI, our method can further improve the performance by acquiring more diversified knowledge from multiple SD HSIs.

B. Effectiveness Analysis of ADSA

The proposed ADSA can align the attributes of heterogeneous HSIs, thus converting them to the same spectral space.

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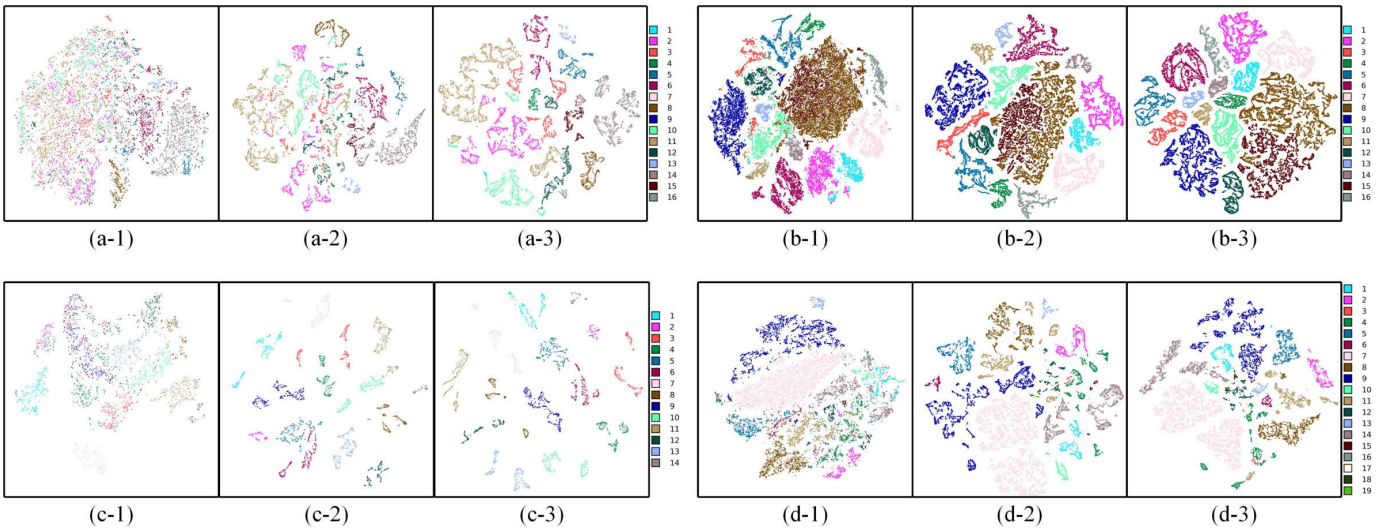


Fig. 8. Visualization of sample distribution using T-SNE, where (-1), (-2), and (-3) denote the distribution maps of original TD samples, TD sample embeddings generated by 3-D RPN trained only on SD samples, and TD sample embeddings after training on both SD and TD samples, respectively. (a-1)–(a-3) Indian Pines. (b-1)–(b-3) Salinas. (c-1)–(c-3) Botswana. (d-1)–(d-3) Chikusei.

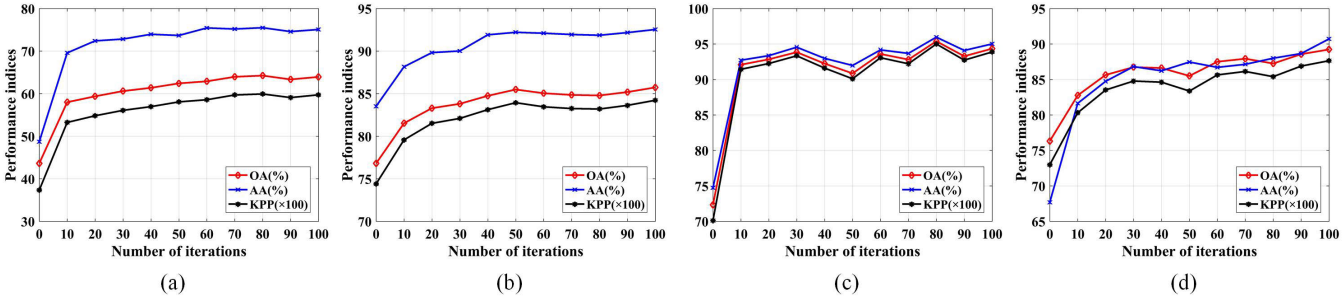


Fig. 9. OA indices with iterations using 3-D RPN trained only on SD samples to directly classify TD samples. (a) Indian Pines. (b) Salinas. (c) Botswana. (d) Chikusei.

TABLE VI
OA INDICES OF OUR METHOD WITH DIFFERENT NUMBERS OF SD HSIS

Dataset	Without SD HSI (Baseline)	With One SD HSI	With Multiple SD HSIs
Indian Pines	66.04±3.81	69.90±2.15	75.63±2.97
Salinas	86.21±2.34	89.45±1.73	91.62±2.51
Botswana	95.44±2.00	96.29±1.55	98.25±0.75
Chikusei	89.81±2.43	90.20±1.58	93.17±1.89

To verify its effectiveness, we replace ADSA with other alignment strategies and compare the performance. Specifically, the alignment strategies for comparison are as follows: 1) randomly sampling 200 bands in each HSI; 2) using PCA to reduce the number of bands in each HSI to 20; and 3) padding each HSI to 204 bands by adding all-zero bands (the maximum number of bands in the four datasets is 204, i.e., Salinas). All the three alignment strategies are used to make the number of bands in different HSIs the same.

As shown in Table VII, alignment using ADSA achieved the highest accuracy on the four different datasets. The random sampling strategy destroys the structures of the spectra, thus degrading the performance on the Indian Pines, Salinas, and Botswana datasets. It indicates that the structural properties of spectra play an important role in knowledge transfer. Similarly, since PCA only extracts the principal components of HSI without considering the structure of the spectra, similar performance degradation occurs on the Botswana. In contrast, although the padding approach does not fully align the attributes of HSIs, it brings improvements on all four datasets, because the raw structures of spectra are not destroyed. When the attributes of HSI are further aligned, our method obtains further improvement, which also proves the effectiveness of the ADSA.

C. Analysis of Feature Transferability

The proposed ADSA-FSL is capable of acquiring discriminative knowledge from the SD HSIs and then applying it to the TD HSI to enhance performance. To investigate this process, we use the t-distributed stochastic neighbor embedding (T-SNE) [45] to visualize the distributions of TD samples in terms of the following: 1) the original TD samples; 2) the TD sample embeddings generated by the 3-D RPN trained on SD samples only; and 3) the TD sample embeddings after training on both SD and TD samples. Note that the distribution map of the second term above does not access any supervised information from the TD HSI, which provides an intuitive

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TABLE VII
OA INDICES OF OUR METHOD WITH DIFFERENT ALIGNMENT STRATEGIES AS WELL AS WITHOUT SD HSI

Dataset	Without SD HSI (Baseline)	With Random Sampling	With PCA	With Padding	With ADSA
Indian Pines	66.04±3.81	64.97±2.10	66.98±1.47	69.81±4.02	75.63±2.97
Salinas	86.21±2.34	82.70±1.51	86.51±2.06	88.43±2.46	91.62±2.51
Botswana	95.44±2.00	95.07±1.67	91.44±2.16	96.61±1.01	98.25±0.75
Chikusei	89.81±2.43	90.49±1.14	90.42±2.03	91.55±2.12	93.17±1.89

TABLE VIII
OA INDICES AND IMPROVEMENTS OF DCFSL AND GIACFSL WHEN IN COMBINATION WITH ADSA

Dataset	DCFSL	ADSA+ DCFSL	Improved	GiaCFSL	ADSA+ GiaCFSL	Improved	CACFSL	ADSA+ CACFSL	Improved
Indian Pines	62.44±4.11	66.77±5.12	4.33% ↑	64.06±4.99	65.84±3.14	1.78% ↑	69.71±3.34	70.49±3.05	0.78% ↑
Salinas	88.96±2.01	89.35±2.13	0.39% ↑	88.30±2.30	88.89±1.70	0.59% ↑	89.57±1.26	90.99±1.21	1.42% ↑
Botswana	96.45±1.01	97.11±0.95	0.66% ↑	96.44±1.10	96.90±1.02	0.46% ↑	96.51±1.01	96.59±1.19	0.08% ↑
Chikusei	90.37±1.91	91.55±1.37	1.18% ↑	88.82±2.20	93.02±1.51	4.20% ↑	92.39±0.64	94.46±0.55	2.07% ↑

TABLE IX
RUNNING TIME OF DIFFERENT METHODS ON EACH DATASET

Dataset	2D-CNN [16]	3D-CNN [17]	SSRN [19]	DFSL [28]	DCFSL [37]	GiaCFSL [38]	CACFSL [39]	Ours
Indian Pines (Testing)	1.2 s	1.3 s	1.8 s	1.2 s	1.2 s	1.2 s	1.3 s	1.9 s
Salinas (Testing)	10.6 s	16.8 s	19.5 s	14.9 s	16.6 s	17.2 s	18.1 s	20.2 s
Botswana (Testing)	0.9 s	1.1 s	1.3 s	1.1 s	1.1 s	1.1 s	1.2 s	1.4 s
Chikusei (Testing)	2.2 s	2.5 s	3.4 s	2.3 s	2.3 s	2.4 s	2.6 s	3.7 s
Average training time	21.5 s	28.8 s	86.7 s	7.6 min	27.7 min	88.5 min	167.8 min	24.1 min

demonstration of the effectiveness of the knowledge learned from the SD datasets on the TD samples.

As shown in Fig. 8, the distribution of raw samples on all four datasets is relatively disorganized, especially the Indian Pines, and thus, directly classifying them leads to undesirable performance, which can also be corroborated by the KNN in Table II. Our approach trains the classification model using SD samples in order to extract transferable knowledge from them. As assumed by meta-learning, the knowledge learned from the SD samples can facilitate the TD classification. As shown in Fig. 8(a-2), (b-2), (c-2), and (d-2), compared with the original sample distributions [shown in Fig. 8(a-1), (b-1), (c-1), and (d-1)], the feature extractor trained only on the SD samples has been able to significantly improve the classification boundaries of the TD samples, where the same-class samples are more clustered together and different-class samples are more distant from each other. This result fully demonstrates that the discriminative knowledge obtained from the SD datasets has an obvious facilitating effect on the TD classification. On this basis, the classification model is further fine-tuned by using a few TD labeled samples, yielding a better classification model on TD HSI, as shown in Fig. 8(a-3), (b-3), (c-3), and (d-3).

To further investigate the contribution of the SD prior knowledge to the TD sample classification, we plot the variation of the TD classification accuracy with the number of iterations. Note that to avoid the interferences from the TD supervised information, in the training process of 3-D RPN, only the SD datasets are available. As shown in Fig. 9, the classification performance of the untrained 3-D RPN model (iteration 0) for the TD is only comparable to that of the KNN, because the classification model at this point does not contain any discriminative knowledge from SD. As the training iterations increase, its classification ability on the TD HSI gradually improves, which indicates that the learned discriminative prior knowledge from the SD samples can effectively improve the distinguishability of the TD samples. These results demonstrate the effectiveness of the proposed method.

D. Generalizability Testing of ADSA

In our method, ADSA is used to obtain distribution-consistent SD samples, from which 3-D RPN is then used to learn transferable knowledge. To verify the generalizability of ADSA, we apply the SD samples generated by ADSA to other similar methods, i.e., DCFSL, GiaCFSL, and CACFSL. For each dataset, we employ ADSA to generate auxiliary samples from the other three HSIs and then use them to test the performances of these three FSL methods, which are referred to as ADSA + DCFSL, ADSA + GiaCFSL, and ADSA + CACFSL, respectively. Each experiment is repeated ten times to report the mean and standard deviation of the OA index.

As shown in Table VIII, with ADSA, the classification performance of DCFSL, GiaCFSL, and CACFSL is significantly improved. Since these three methods use only a single HSI as the SD dataset, the classes and samples of auxiliary dataset are not rich enough, resulting in unsatisfactory performance. ADSA can solve this problem by aligning multiple HSIs with the TD HSI, thus obtaining auxiliary dataset with rich samples and classes. With ADSA, DCFSL achieves a 4.33% improvement in the OA index on the Indian Pines and also gets obvious improvement on the other three datasets. Similarly, GiaCFSL with ADSA shows obvious improvement on all four

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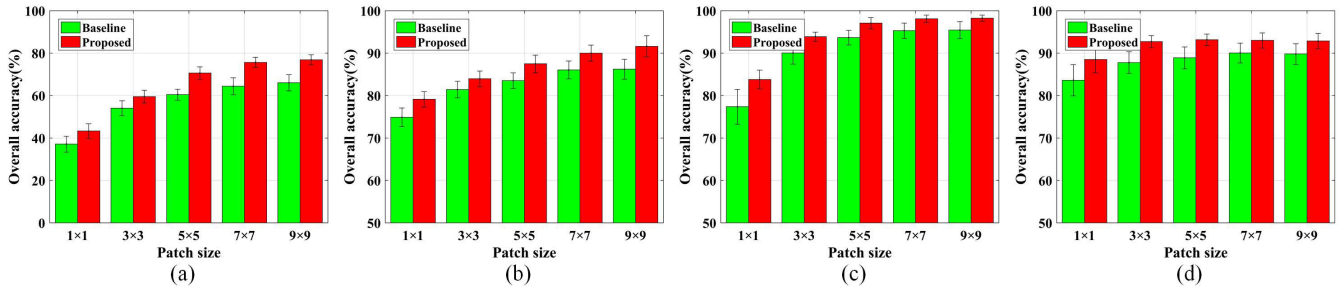


Fig. 10. OA indices of ADSA-FSL and baseline on each HSI using different patch sizes. (a) Indian Pines. (b) Salinas. (c) Botswana. (d) Chikusei.

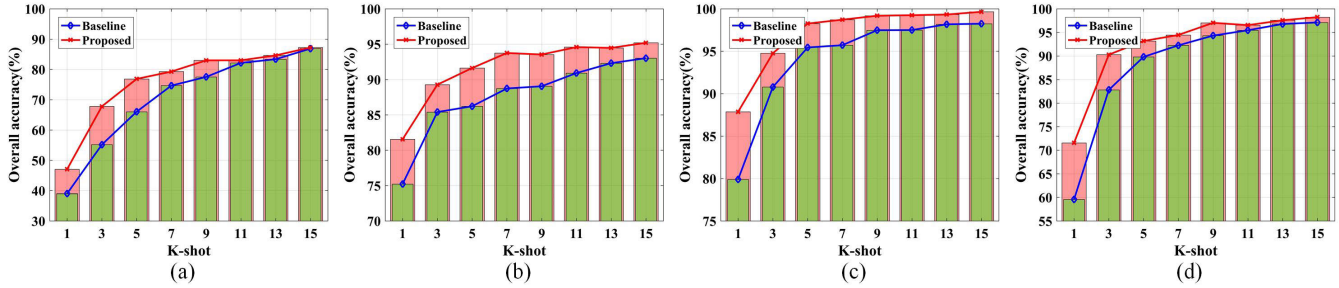


Fig. 11. OA indices of ADSA-FSL and baseline on each HSI using different sample sizes. (a) Indian Pines. (b) Salinas. (c) Botswana. (d) Chikusei.

datasets, especially on the Chikusei, which gains 4.20% OA index improvement. Notably, the classification performance of CACFSL with ADSA on Chikusei even surpasses our method, which once again proves the effectiveness and wide applicability of the proposed ADSA.

E. Analysis of Hyperparameters

Different hyperparameters affect the classification performance of the model. To investigate the proposed method comprehensively, we test the performance of ADSA-FSL with different patch sizes and different numbers of training samples per class and then compare it with the baseline. Each experiment is also repeated ten times to report the mean and standard deviation of the OA index.

In the first set of experiments, we fix the number of training samples per class to 5 (i.e., $K = 5$), and then, set the patch size to $1 \times 1, 3 \times 3, 5 \times 5, 7 \times 7$, and 9×9, respectively, to test the performance of both ADSA-FSL and baseline. As shown in Fig. 10, with the patch size increases, the performance of the model gradually improves. On Botswana and Chikusei, when the patch size exceeds 7×7, the performance improvement is no longer obvious, while the results on Indian Pines and Salinas imply that the performance could be further improved when the patch size is further increased. In addition, even when the patch size is 1×1 (i.e., no spatial information is available), ADSA-FSL still has a stable performance improvement compared with baseline, which indicates that the structure knowledge plays an important role in the cross-domain transfer learning task.

In the second set of experiments, we fixed the patch size to 9×9 and then set K to $[1, 15]$ with an interval of 2, respectively. As shown in Fig. 11, compared with the baseline, the knowledge obtained from the SD HSIs can significantly improve the performance of the TD samples under limited samples. In addition, when K is gradually reduced, our method improves more obviously than the baseline, which is due to the fact that the classification model pretrained on the SD samples already has a certain classification ability on the TD samples, which is consistent with the results in Section V-C. On the contrary, as K increases, the improvement of ADSA-FSL decreases, especially when K reaches 15, with insignificant performance gains on the Indian Pines and Chikusei datasets. It indicates that when there are enough labeled samples in TD HSI, the contribution of the SD knowledge to the TD samples will be reduced, which provides a reference for users to choose the appropriate method for different tasks.

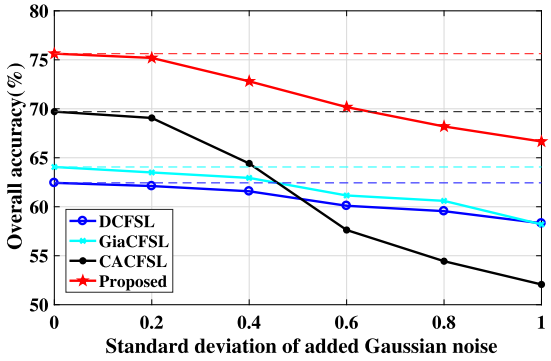


Fig. 12. OA indices of different methods under different noise levels.

F. Robustness to Noise

HSI is often affected by various factors during the imaging process, leading to the presence of noise that can adversely impact classification accuracies. To assess the robustness of the proposed method to noise, we take Indian Pines as an example

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and add Gaussian noise with zero mean and standard deviation ranging from 0 to 1 in the intervals of 0.2 during the training and testing phases. Note that, to prevent noise attenuation during data normalization, a standard normalization process was applied to the HSI before adding noise.

As presented in Fig. 12, with the increased noise level, the OA indices of different methods show a decreasing trend. By observing the gradients of the OA curves, DCFSL demonstrates the best stability, followed by GiaCFSL. Although our method may not have the optimal tolerance to noise, it still achieves the best performance across different noise levels. In addition, compared with CACFSL, the proposed method exhibits acceptable robustness to noise.

G. Running Efficiency

Due to the requirement of a large number of SD samples for auxiliary training, our approach inevitably incurs additional computational overhead. To assess the running efficiency of different methods, we record their testing times on various datasets and report the average training time. As shown in Table IX, compared with traditional deep learning methods, meta-learning-based FSL methods, including DCFSL, GiaCFSL, CACFSL, and our method, all require significantly longer training times. However, once the training is completed, different methods exhibit similar testing times, thus not affecting the deployment and practical usage of the models.

VI. Conclusion

Diverse spectral attributes, such as wavelength and reflectance shift, give rise to distribution gaps between different HSIs, impeding the learning of transferable features. To address this issue, we build the ADSA by embedding spectral attributes into a domain-adaptation model, which can decouple domain-specific attributes from a variety of HSIs in an unsupervised manner. Once the attributes are separated, ADSA can further rebuild new HSIs with identical attributes while maintaining their inherent spectral–spatial structures. Subsequently, the 3-D RPN, combined with an FSL framework, is utilized to model complex HSI structures and learn transferable discriminative knowledge from these attribute-aligned SD HSIs. By applying the learned prior knowledge to the TD HSI, satisfactory performance can be achieved with only a few labeled samples. Our experimental findings illustrate that the proposed ADSA can also enhance other similar methods. In practical HSI imaging, a variety of factors can cause shifts in spectral distribution, such as illumination, capture angles, pixel resolution (mixed spectra), terrain elevation, and resulting shadows. This motivates us to explore further into the embedding of crucial spectral attributes and uncover potential connections between different HSIs in future works, aiming to provide an implementation of generalized large models for intelligent HSI analysis.

References

[1] P. Aastrup, M. P. Tamstorf, and T. Tukiainen, "MINEO. use of hyperspectral data for monitoring pollution from the lead-zinc mine, Mestersvig, in Northeast Greenland," in *Mining in the Arctic*. Boca Raton, FL, USA: CRC Press, 2020, pp. 25–28.

[2] Y. Tan, L. Lu, L. Bruzzone, R. Guan, Z. Chang, and C. Yang, "Hyperspectral band selection for lithologic discrimination and geological mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 471–486, 2020.

[3] C. M. Gevaert, J. Suomalainen, J. Tang, and L. Kooistra, "Generation of spectral–temporal response surfaces by combining multispectral satellite and hyperspectral UAV imagery for precision agriculture applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3140–3146, Jun. 2015.

[4] G. T. Kaya, "A hybrid model for classification of remote sensing images with linear SVM and support vector selection and adaptation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 4, pp. 1988–1997, Aug. 2013.

[5] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[6] H. Yuan and Y. Y. Tang, "Spectral–spatial shared linear regression for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 47, no. 4, pp. 934–945, Apr. 2017.

[7] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[8] S. Huang, H. Zhang, and A. Pizurica, "Subspace clustering for hyperspectral images via dictionary learning with adaptive regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524017.

[9] Y. Gu, Q. Wang, H. Wang, D. You, and Y. Zhang, "Multiple kernel learning via low-rank nonnegative matrix factorization for classification of hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2739–2751, Jun. 2015.

[10] H. Su, Y. Yu, Q. Du, and P. Du, "Ensemble learning for hyperspectral image classification using tangent collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3778–3790, Jun. 2020.

[11] S. Jia, Z. Lin, B. Deng, J. Zhu, and Q. Li, "Cascade superpixel regularized Gabor feature fusion for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 5, pp. 1638–1652, May 2020.

[12] A. Plaza, P. Martinez, R. Perez, and J. Plaza, "A new approach to mixed pixel classification of hyperspectral imagery based on extended morphological profiles," *Pattern Recognit.*, vol. 37, no. 6, pp. 1097–1116, Jun. 2004.

[13] P. Zhong and R. Wang, "Jointly learning the hybrid CRF and MLR model for simultaneous denoising and classification of hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 25, no. 7, pp. 1319–1334, Jul. 2014.

[14] L. Sun, C. He, Y. Zheng, Z. Wu, and B. Jeon, "Tensor cascaded-rank minimization in subspace: A unified regime for hyperspectral image low-level vision," *IEEE Trans. Image Process.*, vol. 32, pp. 100–115, 2023.

[15] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[16] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[17] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[18] S. Jia et al., "A semisupervised Siamese network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5516417.

[19] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[20] W. Wang, S. Dou, Z. Jiang, and L. Sun, "A fast dense spectral–spatial convolution network framework for hyperspectral images classification," *Remote Sens.*, vol. 10, no. 7, p. 1068, Jul. 2018.

[21] X. Zhang et al., "Spectral–spatial self-attention networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5512115.

[22] Q. Liu, L. Xiao, J. Yang, and J. C. Chan, "Content-guided convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6124–6137, Sep. 2020.

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[23] C. Yu, R. Han, M. Song, C. Liu, and C.-I. Chang, "Feedback attention-based dense CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501916.
[24] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.
[25] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, Red Hook, NY, USA, 2017, pp. 4080–4090.
[26] C. Yu, B. Gong, M. Song, E. Zhao, and C.-I. Chang, "Multi-view calibrated prototype learning for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5544713.
[27] D. Pal, V. Bundele, B. Banerjee, and Y. Jeppu, "SPN: Stable prototypical network for few-shot learning-based hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[28] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.
[29] Q. Liu, L. Xiao, N. Huang, and J. Tang, "Composite neighbor-aware convolutional metric networks for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 30, 2023, doi: 10.1109/TNNLS.2022.3232532.
[30] M. Rao, P. Tang, and Z. Zhang, "Spatial–spectral relation network for hyperspectral image classification with limited training samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 5086–5100, Dec. 2019.
[31] J. Zeng, Z. Xue, L. Zhang, Q. Lan, and M. Zhang, "Multistage relation network with dual-metric for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510017.
[32] J. Yang, Y. Zhao, J. C. Chan, and C. Yi, "Hyperspectral image classification using two-channel deep convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5079–5082.
[33] X. He and Y. Chen, "Transferring CNN ensemble for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 876–880, May 2021.
[34] Y. Zhang, W. Li, R. Tao, J. Peng, Q. Du, and Z. Cai, "Cross-scene hyperspectral image classification with discriminative cooperative alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9646–9660, Nov. 2021.
[35] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.
[36] C. Yu, C. Liu, M. Song, and C.-I. Chang, "Unsupervised domain adaptation with content-wise alignment for hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 5511705.
[37] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.
[38] Y. Zhang, W. Li, M. Zhang, S. Wang, R. Tao, and Q. Du, "Graph information aggregation cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 30, 2022, doi: 10.1109/TNNLS.2022.3185795.
[39] W. Wang, F. Liu, J. Liu, and L. Xiao, "Cross-domain few-shot hyperspectral image classification with class-wise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502418.
[40] H. Xu, X. Wang, and J. Ma, "DRF: Disentangled representation for visible and infrared image fusion," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–13, 2021.
[41] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2015, pp. 1180–1189.
[42] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[43] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu, "RoFormer: Enhanced transformer with rotary position embedding," 2021, *arXiv:2104.09864*.
[44] D. Misra, "Mish: A self regularized non-monotonic activation function," in *Proc. Brit. Mach. Vis. Virtual Conf. (BMVC)*, 2020, pp. 1–6.
[45] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, pp. 2579–2605, Nov. 2008.



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